MILITARY SPECIFICATION

BALLISTIC MODIFIER LC-12-15

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for ballistic modifier LC-12-15, a material used in explosive and propellant compositions for military applications (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-313	-	Material Safety Data Sheets, Preparation and Submission of

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-286	-	Propellant, Solid, Sampling, Examination and Testing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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STANDARDS

MILITARY (continued)

MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1171	-	Acceptance and Description Sheets (For Propellants and Explosives)
MIL-STD-1233	-	Procedure for Determining Particle Size, Particle Size Distribution and Packed Density of Powdered Materials
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

(Copies of standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings and publications. The following Government publication forms a part of this specification to the extent specified herein.

49CFR	-	Code of Federal Regulations (Hazardous Materials)

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z129.1	-	American National Standard for the Precautionary Labeling of Hazardous Industrial Chemicals.

(Applications for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

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3.2 Material. The material shall be lead copper resorcylate salicylate (LC-12-15) and may be manufactured according to the following equation.

PbO + 1.5 Salicylic acid + 0.5 beta-resorcylic acid

$$+Cu(OH)_2 \xrightarrow[H_2O]{\text{55 - 60°C (131 - 140°F)}} \text{LC-12-15}$$

The molar ratio of the reactants shall be as shown in the equation, but the total number of moles can be varied to produce different batch sizes. The reactants will be added individually to the reaction vessel and in the order shown. Each reactant will be allowed to react until the solution pH reaches a constant value before the next reactant is added. The reaction will be continued at 55-60°C for a minimum of four (4) hours after all reactants have been added. The LC-12-15 will be removed from solution by filtration and thoroughly washed immediately after completion of the reaction. The LC-12-15 cake will then be dried to constant weight at 77°C, and subsequently ground to break up agglomerates and produce a fine free flowing powder. In addition, the LC-12-15 shall conform to the requirements given in Table I. Deviation from this procedure may be allowed provided that propellant ballistic performance is not affected and prior approval is secured from the Government.

3.3 Chemical and physical properties. The chemical and physical requirements of the material shall conform to those listed in Table I.

3.3.1 Decomposition products. The solution resulting from dissolving LC-12-15 samples as described in paragraph 4.5.3.1 shall be colorless. A yellow coloration indicates the decomposition of beta-resorcylic acid which negates the results of the test in paragraph 4.5.3.1. The test shall then be performed again.

3.4 Material safety data sheets. A material safety data sheet shall be prepared and submitted to the contracting activity, the U.S. Army Environmental Hygiene Agency and the Navy Environmental Health Center in accordance with FED-STD-313. The material shall have no adverse effects on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate medical service who will act as an advisor to the contracting activity (see 4.6 and 6.2(d)).

3.5 Workmanship. The LC-12-15 shall be processed in a manner to assure compliance with all the requirements of this specification. Workmanship shall be such that the material will be uniform, free from contamination, within the limits specified herein and of consistently high quality.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of all examinations and tests of this specification. Approval of the first article samples shall not relieve the supplier of his obligation to meet the quality conformance inspection (4.4). Failure to pass all of the examinations and tests specified herein constitutes cause for rejection of the first article.

4.3.1 First article test samples. Prior to initiation of regular production the contractor shall submit a representative first article sample consisting of approximately 4.5kg(10 lbs). The sample shall be packaged in accordance with instructions issued by the Contracting Officer. All samples submitted shall have been produced by the contractor using the same production process, procedures and equipment as will be used in fulfilling the contract. All materials including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 180 days, or whenever a change occurs in the manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. First article test samples shall be submitted to: Radford Army Ammunition Plant, Radford, VA 24141. The samples shall be plainly identified by an accompanying information sheet with the following:

BALLISTIC MODIFIER, LC-12-15
Samples of Material for First Article Inspection
Name of Manufacturer
Manufacturer's Code No.
Date of Manufacture
Contract No.

4.3.1.1 X-ray diffraction test sample. Immediately prior to initiation of regular production, a 120 gram representative sample of the material shall be submitted to U.S. Army Missile Command, Attn: DRSMI-RKC, Redstone Arsenal, AL 35898 and Naval Ordnance Station, Code 521, Indian Head, MD 20640. This test shall be performed for informational purposes only. The samples shall be plainly identified by an accompanying information sheet with the following:

BALLISTIC MODIFIER, LC-12-15
Samples of Material for X-ray Diffraction Testing
Name of Manufacturer
Manufacturer's Code No.
Date of Manufacture
Contract No.

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4.4 Quality conformance inspection. Quality conformance inspection shall consist of all the tests and examinations of this specification. Failure to pass all of the examinations and tests specified herein constitutes cause for rejection of the lot.

4.4.1 Lot formation. A lot shall consist of one or more batches of LC-12-15 produced by one manufacturer under one continuous set of operating conditions. Each lot shall consist of the quantity of LC-12-15 that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with MIL-STD-105 (or MIL-STD-1235 when applicable). Lot numbering shall be in accordance with MIL-STD-1168.

4.4.2 Sampling. Sampling plans shall be in accordance with MIL-STD-105. Divide the number of containers in the lot into two equal quantities (number of containers). From each part, a composite sample shall be prepared in the following manner: A random sample of filled containers shall be selected in accordance with MIL-STD-105. However, if each part comprises less than five (5) containers, all containers shall be selected. Approximately equal portions shall be taken from the selected containers in such quantities that the aggregate weight of the over-all sample shall be 2.3 kilograms (kg). These portions shall then be thoroughly mixed to form the composite sample. The two composite samples thus formed shall be tested independently in accordance with the tests specified in this specification. A single sample shall be used for the test used to determine particle size. The acceptable quality level (AQL) shall be 0.40 percent for each major defect and 0.65 percent for each minor defect. The classification and code numbers for the tests are given below:

Category	Major Defect
Copper (see 4.5.1)	102
Lead (see 4.5.2)	103
Resorcylic Acid (see 4.5.3)	104
Salicylic Acid (see 4.5.3)	105
Volatiles (see 4.5.4)	106
Acetone Solubility (see 4.5.5)	107
Particle size (see 4.5.6)	108
Detection of Decomposition Products (see 3.3.1)	109

4.4.2.1 Sampling of containers prior to sealing. Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity.

Major Defect (101)	Liner pierced or torn	AQL 0.40%
Minor Defect (201)	Marking missing, incorrect or illegible	AQL 0.65%

4.4.3 Disposition of rejected lots. Upon the rejection of any lot, that rejected lot shall be withdrawn by the contractor and screened for the characteristics that caused the rejection of the lot. Defective batches so found shall be removed from the lot after which the lot may be submitted as an acceptable lot. A failure analysis shall be performed for each defect found

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that exceeds the maximum accept number of the associated sampling plan and corrective action shall be taken to preclude the likelihood of recurrence. A report outlining the results of the failure analysis and the corrective action taken shall be submitted to the procuring activity upon request.

4.5 Test methods.

4.5.1 Test for copper. Weigh accurately to the nearest 0.0001 gram (g), approximately 1.0 gram of the sample into a 250 ml beaker. Add 3 ml concentrated sulfuric acid and 10 ml concentrated nitric acid to the beaker. Cover the beaker with ribbed watch glass and evaporate on a hot plate until white fumes of SO_3 are formed (Caution: Perform in a hood). Cautiously add a few drops of nitric acid by means of a dropper to the beaker without removing the watch glass. Continue heating on the hot plate until evaporation of solution begins and SO_3 fumes are again formed. Repeat addition of nitric acid and heating to SO_3 fumes for 2 or 3 times or until most of the organic matter is destroyed. Cool the beaker and contents, wash cover and sides of beaker with distilled water. Add 100 ml of 50% ethyl alcohol to the beaker and allow to stand at 50-60°C for 30 minutes. Filter contents of the beaker through a Selas medium porosity or 3010 crucible using additional 50% ethyl alcohol to aid in the quantitative transfer of this solution. Collect the filtrate in a 400 ml beaker. Evaporate contents of the beaker (covered with a ribbed watch glass to prevent loss from spattering) until SO_3 fumes form. Quantitatively transfer the contents of the beaker to a 250 ml beaker with the aid of distilled water. Dilute the contents of the beaker to about 150 ml with distilled water and add 10 ml concentrated ammonium hydroxide with stirring. If this solution turns black or dark blue, add concentrated sulfuric acid dropwise until the color of the solution becomes light colored. Add 11 ml concentrated nitric acid, dilute to 200 ml with distilled water and electrolyze at 1.2-2.0 amperes for an hour, using a weighed platinum gauze cathode and a rotating anode. Rinse, dry and weigh the cathode.

Calculation of % Copper:

$$\% \text{ Copper} = \frac{\text{Weight of copper deposited} \times 100}{\text{Sample weight on a Volatile Free Basis}}$$

4.5.1.1 Alternate test for copper. The copper content may be determined using an Atomic Absorption Method which appears in a standard or reference textbook.

4.5.2 Test for lead. Weigh accurately to the nearest 0.0001 gram(g), approximately 0.3g of sample and perform the same test procedure as that used for the detection of copper (see 4.5.1) until the point of filtration. Using acetone as a washing medium, filter the contents of the beaker through a Selas medium porosity or 3010 crucible. Dry the crucible and contents at approximately 600°C (1112°F) for 20 minutes, cool and weigh.

Calculation of % Lead:

$$\% \text{ Lead} = \frac{\text{Weight of precipitate} \times 68.32}{\text{Weight of sample on a volatile free basis}}$$

4.5.2.1 Alternate test for lead. The lead content may be determined using an Atomic Absorption Method which appears in a standard or reference textbook.

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4.5.3 Test for resorcylic and salicylic acids. The analysis of these acids shall be run concurrently.

4.5.3.1 Test for resorcylic acid. In separate 250 ml beakers, place accurately weighed portions to the nearest 0.0001 grams of approximately 0.04 and 0.08 grams of beta-resorcylic acid and 0.15 grams of the sample. Add 100 ml of distilled water and 1.0 ml of concentrated hydrochloric acid by pipette to each. Warm the beakers on a steam plate or preheated hot plate. Do not allow the solution temperature to exceed 65°C (149°F) or the total heating time to exceed 30 minutes. These heating precautions are necessary to prevent decomposition of beta-resorcylic acid. Stir the contents of the beakers with a glass stirring rod to dissolve the solids. Cool the contents of the beakers and quantitatively transfer the contents of each beaker to a separate 250 ml volumetric flask using distilled water to aid in the transfer. Dilute to 250 ml with distilled water then pipette a 3 ml portion of each solution into a separate 100 ml volumetric flask and dilute to 100 ml with distilled water. Allow these solutions to stand for one hour before measuring absorbances. If the solution has a yellow discoloration the solution shall be discarded and the test run again(see 3.3.1). Determine absorbances of the diluted solutions in a 1 cm cell on an ultraviolet spectrophotometer at 254 nanometers (nm) and 235 nm, using distilled water as the blank. The resorcylic acid standard can not be saved for reuse because it is unstable.

4.5.3.1.1 Calculation of % resorcylic acid. Dilutions need not be considered in calculations because sample and standards are diluted alike. For each wavelength, calculate absorbance per gram of resorcylic acid (APGres) and absorbance per gram of sample (APGsmpl) by the relationship:

$$APG = \frac{\text{Absorbance}}{\text{Weight on a volatile free basis}}$$

Calculate the average (APGres) for each wavelength and determine as follows:

Weight percentage of beta-resorcylic acid:

$$\frac{100 \times APG^{235}_{sal} \times APG^{254}_{smpl} - 100 \times APG^{235}_{smpl} \times APG^{254}_{sal}}{APG^{235}_{sal} \times APG^{254}_{res} - APG^{235}_{res} \times APG^{254}_{sal}}$$

4.5.3.2 Test for salicylic acid. Determine the percentage of salicylic acid in the sample by using the procedure described in 4.5.3.1 except that 0.04 and 0.08 grams of salicylic acid shall be substituted for the resorcylic acid.

4.5.3.2.1 Calculation of % salicylic acid.

Weight percentage of salicylic acid:

$$\frac{100 \times APG^{235}_{smpl} - APG^{235}_{res} \times \%\ \text{res acid}}{APG^{235}_{sal}} - 0.8$$

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4.5.4 Test for percent volatiles. Dry a 2 to 4 gram sample of LC-12-15 at 105-110°C (221-230°F) for one hour. Determine % volatiles according to Method 101.2.2 of MIL-STD-286.

4.5.5 Test for acetone solubility. Place a weighed sample of approximately 0.1 ± 0.0001 gram in a tared fine porosity glass crucible. Place crucible in a 250 ml beaker and add acetone to fill half the crucible. Replenish the acetone, as necessary, keeping the sample covered with acetone for 30 minutes. Filter the contents of the crucible with suction and wash contents of the crucible with 75 ml of acetone. Dry the crucible and contents in an oven at 100 ± 5°C for 1 hour, cool and weigh.

Calculation of % acetone-solubility:

$$\% \text{ acetone-solubility} = \frac{\text{loss in weight} \times 100}{\text{sample weight}} - \% \text{ volatiles (see 4.5.4)}$$

4.5.6 Test for particle size. Determine the average particle size of the sample using Method 100 of MIL-STD-1233 with a sample of 2.765 ± 0.001 grams and porosity setting of 0.51.

4.6 Submission of material safety data sheets. Material Safety Data Sheets prepared as specified in 3.4 shall be submitted by the manufacturer. One copy shall accompany the samples being submitted for testing.

4.7 Packaging inspection.

4.7.1 Examination of packaging and marking. An examination shall be made to determine that preservation, packing and marking comply with the requirements of Section 5 of this specification. Defects shall be scored in accordance with the list following. The sample unit for this examination shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The samples for this examination shall be selected at random in accordance with MIL-STD-105, inspection level S-2 and acceptable quality level (AQL) 4.0 defects per hundred units.

Examine	Defect
Packaging	Container not as specified, closures not accomplished by specified or required methods or materials. Leakage or seepage of contents. Non-conforming component, component missing, damaged or otherwise defective. Bulged, distorted or broken container.
Markings	Data, including directions for use, omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements.

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5. PACKAGING

5.1 Preservation and Unit Packing. The ballistic modifier LC-12-15 shall, in the quantity specified in the contract or order, be packaged and marked in accordance with the requirements of Title 49 of the Code of Federal Regulations.

5.2 Marking. Containers shall be marked in accordance with MIL-STD-129. Markings shall include, but are not limited to the following:

a. Manufacturer's Name
b. Product designation
c. Lot number
d. Date of manufacture
e. Number of this specification

In addition to the markings required above, unit pack markings shall contain use, mixing, handling and precautionary markings as well as noting any protective clothing and equipment required. All unit and intermediate packs of toxic and hazardous chemicals and materials shall also be labeled in accordance with the applicable laws, statutes, regulations or ordnances, including Federal, State, and municipal requirements. In addition, unit and intermediate containers, including unit containers that serve as shipping containers such as pails and drums, shall be marked with the applicable precautionary information detailed in American National Standard ANSI Z129.1.

6. NOTES

6.1 Intended use. Ballistic modifier LC-12-15 is intended for use in explosive and propellant compositions for military applications.

6.2 Ordering data. Requisition, contracts or orders should specify the following:

a. Title, number and date of this specification.
b. Acceptance and description sheets for each lot in accordance with MIL-STD-1171, if specified.
c. Provisions for submission of first article samples.
d. Specify FAR Clauses 52.223-3, 23.301 and 23.302.
e. Quantity

6.2.1 First article inspection. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations.

Custodians:
Army - AL
Navy - AS

Preparing activity:
Navy - AS
(Project No. 6810-B423)

Review activities:
Army - MI
Navy - OS
Air Force - 12, 18

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Table I. Chemical and physical properties.

Property	Min	Max	Test paragraph
Copper, Weight %	10.1	12.9	4.5.1
Salicylic Acid, Weight %	34.5	39.7	4.5.3.2
Resorcylic Acid, Weight %	11.0	16.6	4.5.3.1
Lead, Weight %	35.8	39.4	4.5.2
Volatiles, Weight %	-	4.0	4.5.4
Acetone Solubility, Weight %	-	2.0	4.5.5
Particle Size, micrometers, µm	-	8.0	4.5.6

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-85735	BALLISTIC MODIFIER LC-12-15

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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